

FlowRoute: Small, Example-Aware Workflow Routing Without Generative Inference

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September 2026

Abstract

Natural-language interfaces often send every request to a generative language model even when the available actions are fixed workflows. We study a smaller alternative: encode the request, each workflow description, and a few approved examples with a frozen 22.7M-parameter sentence encoder, then route by cosine similarity or abstain. On the public CLINC150 and BANKING77 test sets, represented as catalogs of 150 and 77 workflows, ten examples per workflow yield $85.01\% \pm 0.45$ and $83.10\% \pm 0.54$ top-1 accuracy over five example-selection seeds. These results improve on description-only dense routing by 19.64 and 21.77 percentage points, respectively. Recall@5 reaches 98.00% and 96.88%. On CLINC150, validation-calibrated abstention provides $64.01\% \pm 2.86$ in-scope coverage with $4.60\% \pm 0.99$ selective error and a $2.92\% \pm 0.63$ out-of-scope false-positive rate. Single-thread CPU latency has a 13.48 ms p95 for a 150-workflow catalog. The results show that small, non-generative routers are practical when workflows include representative examples, while also showing that confidence thresholds are not execution authorization.

1 Introduction

Many application requests correspond to deterministic operations such as checking a balance, updating a reminder, or cancelling a transfer. A generative large language model (LLM) can select those operations, but it also adds cost, latency, stochastic output, and a broader security boundary. Fixed-label intent classifiers avoid generation but assume that the action set is known during training. In practice, workflow catalogs vary by tenant and change over time.

We examine a simple runtime-catalog setting. Each workflow supplies a short description and zero or more approved example requests. A compact sentence encoder embeds these texts. The router selects the highest-scoring workflow only when both its score and its margin over the runner-up clear validation-selected thresholds; otherwise it returns a fallback decision. Candidate embeddings are cached, and the router never generates text or executes an action.

This paper makes three contributions. First, it gives a reproducible example-aware scoring and abstention rule for changing workflow catalogs. Second, it reports real measurements on two public intent datasets converted into runtime catalogs, including five example-selection seeds and an untouched test

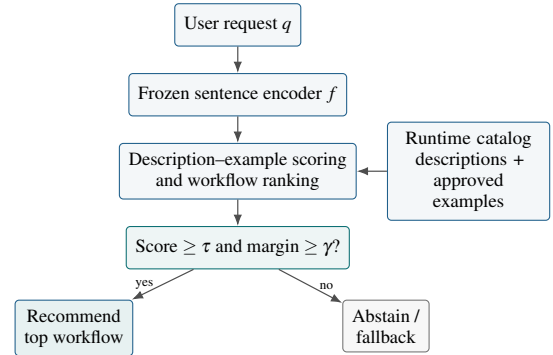


Figure 1: FlowRoute’s online decision path. Catalog embeddings are cached; the request is encoded once, scored against runtime workflow evidence, and either recommended or rejected by the calibrated abstention gate. Execution remains subject to independent policy checks.

set. Third, it quantifies the effect of adding workflow examples: ten examples improve top-1 accuracy by 19.64–21.77 points over description-only routing. The contribution is a compact empirical system study, not a new pretrained language model or a claim of state-of-the-art tool use.

Tool retrieval systems such as ToolRerank and SkillRouter retrieve relevant APIs or skill documents for downstream LLM agents [8, 7]. Our scope is narrower: choose one deterministic workflow or abstain before execution. Sentence-BERT made independent sentence encoding practical for semantic retrieval [3], and MiniLM reduced transformer size through distillation [5]. We use an existing MiniLM sentence encoder and focus on catalog evidence and the final decision boundary.

2 Method

Figure 1 summarizes the online path. Let q be a request and $\mathcal{W} = \{w_1, \dots, w_n\}$ a runtime catalog. Workflow w_i has description d_i and approved examples E_i . A frozen encoder f maps each text to a unit vector. We score each workflow by

$$s_i(q) = \alpha f(q)^\top f(d_i) + (1 - \alpha) \max_{e \in E_i} f(q)^\top f(e), \quad (1)$$

with the second term omitted when E_i is empty. The maximum lets one close example provide strong evidence without concatenating the entire contract into a single truncated input. Description and example embeddings are computed when

the catalog is loaded; online inference requires one request encoding and matrix products.

The router predicts $i^* = \arg \max_i s_i$. Let $s_{(1)}$ and $s_{(2)}$ be the two largest scores. It returns $\text{route}(i^*)$ only if

$$s_{(1)} \geq \tau \quad \text{and} \quad s_{(1)} - s_{(2)} \geq \gamma; \quad (2)$$

otherwise it returns FALLBACK. The mixing weight α and thresholds (τ, γ) are selected on validation data only.

This output is a recommendation, not execution authority. A workflow engine must still enforce authentication, typed inputs, authorization, confirmation, idempotency, and current workflow availability. High-impact writes should require independent policy checks even when Eq. 2 accepts the route.

3 Experimental Setup

Datasets. CLINC150 contains 150 in-scope intents with 15,000 training, 3,000 validation, and 4,500 test requests, plus dedicated out-of-scope (OOS) examples [2]. We use its 100 OOS training and 100 OOS validation requests for threshold selection and its 1,000 OOS test requests only for final evaluation. BANKING77 contains 13,083 requests across 77 fine-grained banking intents, split into 10,003 training and 3,080 test examples [1]. Because it has no official validation split, we deterministically reserve 20 training requests per intent (1,540 total) before sampling catalog examples.

Each intent becomes one workflow. Its description is generated by replacing underscores in the label with spaces and placing the result in the fixed template “Workflow for [label].” We sample $k \in \{0, 1, 3, 5, 10\}$ examples per workflow from the remaining training data using seeds 11, 23, 37, 53, and 71. No task-specific gradient update is performed.

Model and baselines. All dense runs use the 22.7M-parameter all-MiniLM-L6-v2 encoder at revision 1110a243; it maps text to 384 dimensions and contains exactly 22,713,216 parameters [4]. We compare (i) dense description-only routing, (ii) a lexical baseline combining word 1–2 gram TF-IDF and character 3–5 gram TF-IDF with weights 0.65 and 0.35, and (iii) Eq. 1. The lexical baseline uses the same ten examples per workflow.

For each seed and example budget, α is selected from $\{0, 0.1, \dots, 1\}$ to maximize validation top-1 accuracy, preferring more description weight on ties. We report mean and sample standard deviation across the five example selections. The test sets are not used for model, weight, or threshold selection.

Selective routing. On CLINC150, we search τ in increments of 0.005 and γ in increments of 0.005, maximizing in-scope validation coverage. The upper endpoints of two-sided 95% Wilson intervals [6] for both selective error among accepted in-scope routes and OOS false-positive rate must be at most 5%. At least 100 validation routes must be accepted. This is a finite-sample selection rule, not a distribution-free deployment guarantee.

Efficiency. Latency is measured for 500 batch-size-one requests after 50 warm-up requests, with one PyTorch CPU thread on an AMD EPYC 9V74. The 150 descriptions and 1,500 example embeddings are cached. Timing includes tokenization, request encoding, scoring, and selection, but excludes startup and catalog encoding.

4 Results

Ranking. Table 1 shows a monotonic mean improvement as more workflow examples are added in the tested range. Ten examples improve top-1 accuracy over description-only dense routing by 19.64 points on CLINC150 and 21.77 points on BANKING77. The same method exceeds the ten-example lexical baseline by 4.31 and 10.75 points. Recall@5 of 98.00% and 96.88% indicates that a later verifier could inspect a small candidate set even when top-1 routing fails.

Abstention. The validation rule accepts 64.01% of in-scope test requests on average while keeping mean selective error at 4.60% and OOS false positives at 2.92% (Table 2). One of five seeds reaches 5.47% selective error on the test set despite satisfying the validation confidence-bound constraint. Threshold calibration therefore reduces error but does not by itself guarantee a production risk limit under sampling variation or distribution shift.

Efficiency. The batch-size-one latency median is 11.54 ms; p95 and p99 are 13.48 and 14.88 ms. These measurements include the online path described above and use one CPU thread. The encoder has 22.7M parameters. Results will vary by processor, runtime, catalog size, and request length.

Error analysis. For the reference seed 37, common CLINC150 confusions are semantically adjacent pairs such as time/timezone, update-versus-read variants of lists and reminders, and user-name versus assistant-name changes. BANKING77 errors similarly cluster around reverted versus pending top-ups, reasons for identity verification versus the verification action, and bank-transfer charges versus transfer fees. These failures support using explicit exclusions, typed state, or a pairwise verifier before consequential execution.

5 Limitations and Safety

Intent labels and historical requests are only proxies for real workflow contracts. Both datasets are English, single-intent, and customer-assistant oriented; they do not test multi-step composition, adversarial instructions, multilingual traffic, catalog updates, or authorization. The workflow descriptions are mechanically derived from labels, so manually reviewed capability and exclusion text may change results. Conversely, benchmark examples may be cleaner than production traffic. Pretraining overlap for the public encoder cannot be ruled out.

The experiment compares compact non-generative methods and does not establish superiority over cross-encoders or LLM

Table 1: Test ranking results in percent. Values are mean $\pm$ sample standard deviation across five workflow-example selections. Description-only routing is deterministic; its standard deviation is zero. Best values are bold.

System	Examples	CLINC150 (150 workflows)		BANKING77 (77 workflows)	
		Top-1	Recall@5	Top-1	Recall@5
Dense description only	0	65.38	88.47	61.33	83.90
Word/character TF-IDF	10	80.71 $\pm$ 0.35	94.75 $\pm$ 0.43	72.35 $\pm$ 1.23	93.44 $\pm$ 0.39
FLOWROUTE	1	74.41 $\pm$ 1.33	94.28 $\pm$ 0.61	67.56 $\pm$ 0.82	89.92 $\pm$ 0.76
FLOWROUTE	3	79.89 $\pm$ 0.53	96.66 $\pm$ 0.45	73.68 $\pm$ 0.48	93.70 $\pm$ 0.33
FLOWROUTE	5	81.89 $\pm$ 0.76	97.32 $\pm$ 0.31	78.34 $\pm$ 0.68	95.28 $\pm$ 0.31
FLOWROUTE	10	85.01 $\pm$ 0.45	98.00 $\pm$ 0.13	83.10 $\pm$ 0.54	96.88 $\pm$ 0.14

Table 2: CLINC150 selective-routing test results in percent for ten examples per workflow.

Metric	Mean $\pm$ SD	Seed range
In-scope coverage	64.01 $\pm$ 2.86	59.71–67.71
Selective error	4.60 $\pm$ 0.99	3.13–5.47
OOS false-positive rate	2.92 $\pm$ 0.63	2.30–3.70

routers. Five example selections measure sensitivity to catalog evidence, not uncertainty over new test populations. OOS calibration uses only 200 examples, and its Wilson rule assumes that validation is relevant to deployment. A production system needs domain-specific data, continuous monitoring, conservative fallback, and independent execution policy. It must not treat similarity or confidence as permission to send messages, move money, delete data, or change access.

The benchmark script records package versions, model revision, data URLs, SHA-256 checksums, seeds, every threshold, and complete per-seed results. The accompanying source archive excludes raw datasets and downloads them from their official repositories under their original licenses: CC BY 3.0 for CLINC150, CC BY 4.0 for BANKING77, and Apache 2.0 for the encoder.

Code availability. The open-source FlowRoute implementation, production integration examples, tests, and developer documentation are available at <https://github.com/open-first/FlowRoute>. The arXiv source archive contains the exact benchmark script and result files used in this paper.

6 Conclusion

A small frozen encoder can route requests to a changing catalog without generative inference. On two public datasets, representative workflow examples provide larger gains than descriptions alone, and ten examples reach 83–85% top-1 accuracy with roughly 14 ms p95 single-thread CPU latency. Calibrated abstention lowers error at the cost of coverage, but the observed seed variation reinforces a strict boundary between route recommendation and execution authority. The practical next step is evaluation on versioned, side-effect-aware workflow contracts from real deployments.

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